

Renewable Energy Grid Stability & Battery Storage

Draft 2

1 Introduction

The increasing use of renewable energy introduces uncertainty into electricity generation. Unlike conventional generation, wind and solar power are dependent on environmental conditions and therefore vary over time. At the same time, electricity consumption varies according to time of day, day of the week, weather, and season.

This creates a fundamental supply–demand problem:

$$\text{Generation} + \text{Storage} \geq \text{Demand}. \quad (1)$$

When available generation and battery supply are insufficient to meet demand, a loss of load occurs.

This project develops a probabilistic reference model for a renewable-energy system containing wind generation, solar generation, consumer demand, and battery storage. The objective is to determine the probability that the system is unable to satisfy demand.

2 Research Question

2.1 Main Research Question

To what extent can battery storage reduce the probability of loss of load in a grid with stochastic renewable generation and consumer demand?

2.2 Supporting Questions

1. How can wind, solar generation, and consumer demand be represented using appropriate probabilistic models?
2. How does battery capacity and state-of-charge affect the probability of an energy deficit?
3. Does load-shifting reduce peak demand and improve grid reliability?

3 Background and Motivation

3.1 Wind

Wind speed is a continuous and highly variable quantity. The Weibull distribution is one of the most commonly used probability distributions for modelling wind speed, although other distributions can provide better fits for particular locations or time periods.

The reference model assumes:

$$W_t \sim \text{Weibull}(k, \lambda), \quad (2)$$

where k is the shape parameter and λ is the scale parameter.

Its probability density function is

$$f_W(w) = \frac{k}{\lambda} \left(\frac{w}{\lambda} \right)^{k-1} e^{-(w/\lambda)^k}, \quad w \geq 0. \quad (3)$$

The corresponding cumulative distribution function is

$$F_W(w) = P(W_t \leq w) = 1 - e^{-(w/\lambda)^k}. \quad (4)$$

This allows the model to associate a probability with different wind-speed ranges.

However, wind speed is not itself electrical generation. A turbine power function is therefore required to transform wind speed into power:

$$P_t^{\text{wind}} = f_{\text{wind}}(W_t). \quad (5)$$

A simplified model may use a cubic relationship between cut-in and rated wind speeds:

$$P_t^{\text{wind}} = P_r \left(\frac{W_t - W_{ci}}{W_r - W_{ci}} \right)^3, \quad W_{ci} \leq W_t < W_r, \quad (6)$$

with zero generation below the cut-in speed and a maximum output of P_r above the rated speed.

3.2 Solar

Solar irradiance also varies over time and has strong daily and seasonal patterns. Unlike wind, solar generation has a natural day–night cycle. Therefore, solar generation should be treated differently during daylight and nighttime.

For normalized daylight irradiance $S_t \in [0, 1]$, a Beta distribution can be used:

$$S_t \sim \text{Beta}(\alpha, \beta). \quad (7)$$

Its probability density function is

$$f_S(s) = \frac{s^{\alpha-1}(1-s)^{\beta-1}}{B(\alpha, \beta)}, \quad 0 \leq s \leq 1. \quad (8)$$

The normalized irradiance can then be converted to actual irradiance using

$$I_t = S_t I_{\text{max}}. \quad (9)$$

Solar power is subsequently obtained through a photovoltaic conversion function:

$$P_t^{\text{solar}} = f_{\text{solar}}(I_t). \quad (10)$$

At nighttime,

$$P_t^{\text{solar}} = 0. \quad (11)$$

Therefore, a simplified solar model is

$$P_t^{\text{solar}} = \begin{cases} 0, & t \in \text{night}, \\ f_{\text{solar}}(I_t), & t \in \text{day}. \end{cases} \quad (12)$$

This allows the model to incorporate both the probability distribution of daylight irradiance and the deterministic day–night cycle.

4 Reference Renewable Generation Model

The total renewable generation at time t is the sum of wind and solar generation:

$$\boxed{G_t = P_t^{\text{wind}} + P_t^{\text{solar}}} \quad (13)$$

Thus, the complete renewable-generation process is

$$W_t \rightarrow P_t^{\text{wind}} \quad (14)$$

and

$$S_t \rightarrow I_t \rightarrow P_t^{\text{solar}}. \quad (15)$$

This distinction is important because wind speed and solar irradiance are environmental variables, whereas G_t represents electrical generation.

5 Consumer Demand

Electricity demand is also stochastic but exhibits strong temporal structure. Consumption generally varies throughout the day and may differ between weekdays and weekends and across seasons.

A reference model must therefore specify a probability distribution for consumer demand:

$$D_t \sim F_D(\theta_t), \quad (16)$$

where θ_t represents parameters that may depend on time of day, day type, or season.

A simple initial reference model is a time-dependent Normal distribution:

$$D_t \sim \mathcal{N}(\mu_t, \sigma_t^2), \quad (17)$$

where

$$\mu_t = f(\text{hour, day type, season}). \quad (18)$$

Alternative demand models, such as Lognormal, Gamma, empirical, or Markov-based models, can later be compared with this reference model.

6 Battery Storage Model

Battery storage introduces a state-dependent component into the system. The amount of energy stored in the battery and the amount of power that can be supplied during a particular hour are distinct quantities.

Let

$$SOC_t \quad (19)$$

represent the energy stored in the battery at time t , with capacity C :

$$0 \leq SOC_t \leq C. \quad (20)$$

The battery state evolves according to charging and discharging:

$$SOC_{t+1} = SOC_t + \eta_c P_t^{\text{ch}} \Delta t - \frac{P_t^{\text{dis}} \Delta t}{\eta_d}, \quad (21)$$

where:

- P_t^{ch} is charging power,

- P_t^{dis} is discharging power,
- η_c is charging efficiency,
- η_d is discharging efficiency,
- Δt is the time interval.

If renewable generation exceeds demand,

$$G_t > D_t, \quad (22)$$

the surplus may be used to charge the battery.

If

$$G_t < D_t, \quad (23)$$

the battery may discharge to supply part of the deficit.

The usable battery output is therefore constrained by both the stored energy and the maximum discharge rate:

$$B_t = P_t^{\text{dis}} \leq \min \left(P_{\text{dis}}^{\text{max}}, \frac{\eta_d \text{SOC}_t}{\Delta t} \right). \quad (24)$$

6.1 Markov Chain Representation

The battery can be represented as a discrete Markov chain by dividing its state-of-charge into discrete states.

For battery states i and j ,

$$P(\text{SOC}_{t+1} = j \mid \text{SOC}_t = i) \quad (25)$$

represents the probability of transitioning from state i to state j .

The Markov assumption is that the next battery state depends primarily on the current state and the conditions during the current time interval, rather than the complete history of previous states.

7 Energy Balance

The central random variable of the model is the net energy balance:

$$\boxed{E_t = G_t + B_t - D_t} \quad (26)$$

where

- G_t is renewable generation,
- B_t is usable battery output,
- D_t is consumer demand.

Substituting the renewable-generation model gives

$$\boxed{E_t = P_t^{\text{wind}} + P_t^{\text{solar}} + B_t - D_t.} \quad (27)$$

The sign of E_t determines whether the system can satisfy demand:

$$E_t \geq 0 \quad \Rightarrow \quad \text{demand can be satisfied}, \quad (28)$$

while

$$E_t < 0 \quad \Rightarrow \quad \text{energy deficit.} \quad (29)$$

The magnitude of the deficit is

$$\text{Deficit}_t = \max(0, -E_t). \quad (30)$$

8 Loss-of-Load Probability

Define an indicator random variable:

$$I_t = \begin{cases} 1, & E_t < 0, \\ 0, & E_t \geq 0. \end{cases} \quad (31)$$

Then the probability of loss of load is

$$P(I_t = 1) = P(E_t < 0) \quad (32)$$

and the Loss-of-Load Probability (LOLP) is

$$LOLP = P(E_t < 0). \quad (33)$$

For T simulated or observed time intervals, the empirical estimate is

$$\widehat{LOLP} = \frac{1}{T} \sum_{t=1}^T I_t. \quad (34)$$

9 Distribution of the Energy Balance

The central mathematical problem is to determine the distribution of

$$E_t = P_t^{\text{wind}} + P_t^{\text{solar}} + B_t - D_t. \quad (35)$$

Although the individual components may have known distributions, the resulting distribution of E_t does not generally correspond to a standard named distribution.

In particular, the system contains:

- a Weibull-distributed wind variable,
- a Beta-distributed normalized solar variable,
- a demand distribution,
- a state-dependent battery process.

The distribution of E_t therefore depends on the chosen models and on the relationships between the variables.

The reference model will use simulation to obtain the empirical distribution of E_t and estimate

$$P(E_t < 0). \quad (36)$$

Different choices of probability models can then be compared:

Each choice produces a potentially different distribution for E_t and therefore a different estimate of LOLP.

Model	Wind	Solar	Demand
Reference	Weibull	Beta	Time-dependent Normal
Alternative 1	Weibull	Beta	Lognormal
Alternative 2	Weibull	Beta	Gamma
Alternative 3	Weibull	Beta	Empirical/Markov

Table 1: Candidate probabilistic models for the system.

10 Statistical Analysis

Once the reference model has been constructed, additional probability and statistical methods can be applied.

10.1 Central Limit Theorem

The Central Limit Theorem can be used to investigate the behaviour of aggregate renewable generation over multiple observations.

10.2 Chebyshev’s Inequality

Chebyshev’s inequality provides a distribution-independent bound:

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}. \quad (37)$$

This can be used to establish bounds on unusually large deviations in the model.

10.3 Hypothesis Testing

To investigate load-shifting, peak demand can be compared before and after demand shifting.

For example:

$$H_0 : \mu_{\text{peak,shifted}} = \mu_{\text{peak,original}} \quad (38)$$

against

$$H_1 : \mu_{\text{peak,shifted}} < \mu_{\text{peak,original}}. \quad (39)$$

11 Assumptions and Predictions

The reference model will make the following assumptions:

1. Wind speed can be approximated using a Weibull distribution.
2. Normalized daylight solar irradiance can be approximated using a Beta distribution.
3. Solar generation is approximately zero during nighttime.
4. Consumer demand follows a time-dependent probability distribution.
5. Battery state-of-charge is finite and evolves according to charging and discharging.
6. Battery output is limited by both its available stored energy and maximum discharge rate.
7. The system is evaluated using hourly time intervals.

The primary prediction is that increasing battery capacity will reduce the probability of an energy deficit:

$$C \uparrow \Rightarrow P(E_t < 0) \downarrow. \quad (40)$$

However, the reduction in LOLP is expected to exhibit diminishing returns once the battery is sufficiently large relative to typical generation deficits.

12 Planned Approach

The project will proceed through the following stages:

1. Establish probability models for wind speed, solar irradiance, and consumer demand.
2. Define the functions converting environmental variables into renewable power generation.
3. Construct the battery charging, discharging, and state-transition model.
4. Combine the components into the energy-balance variable

$$E_t = G_t + B_t - D_t.$$

5. Determine the empirical distribution of E_t using simulation.
6. Estimate

$$LOLP = P(E_t < 0).$$

7. Compare LOLP for different battery capacities.
8. Investigate whether load-shifting changes peak demand and system reliability.
9. Compare alternative modelling assumptions and their effects on the final distribution of E_t .

13 Scope

This project focuses on probabilistic supply–demand reliability. It does not attempt to model the complete physical stability of a power grid, including voltage stability, frequency dynamics, transmission constraints, or generator faults.